From P-Values to Simple Linear Regression

ISyE 4031, Week 2 Class 3

Prof. Shihao Yang
H. Milton Stewart School of Industrial and Systems Engineering
Georgia Institute of Technology

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Today: close inference, then build a regression line

By the end of class, you should be able to:

1. interpret a p-value and choose the correct rejection region;
2. explain Type I error, Type II error, and power;
3. write a simple linear regression model and identify its parts;
4. derive and interpret the least-squares line;
5. audit a fitted line using residual identities and a hand calculation.

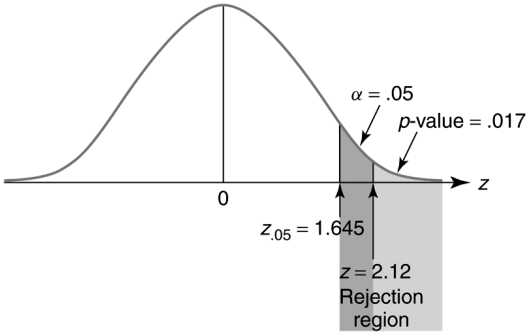
Inference asks whether a signal is credible; regression measures that signal.

A p-value measures compatibility with the null model

Assuming H_0, the p-value is the probability of a statistic at least as extreme as the observed statistic.

It is **not**:

- $P(H_0 \text{ is true} \mid \text{data})$;
- the probability the result occurred by chance;
- the size or importance of an effect.



A p-value does not make the decision for us

Suppose a two-sided test gives

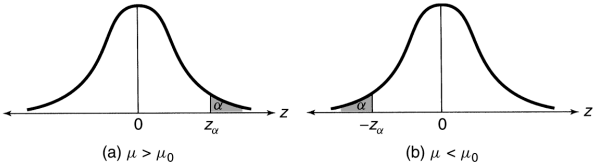
$$t_{\text{obs}} = 2.31, \qquad df = 18, \qquad p = 0.033.$$

$\alpha = 0.05$
Reject H_0 because $0.033 < 0.05$.

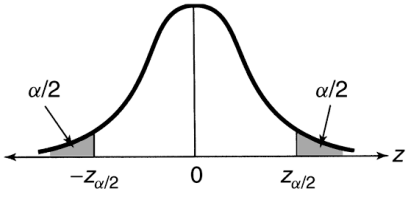
$\alpha = 0.01$
Do not reject H_0 because $0.033 > 0.01$.

Report the effect, uncertainty, assumptions, and decision threshold.

The alternative determines the rejection region



One-sided alternatives



Two-sided alternative

Choose the direction before looking at the data.

Type I error, Type II error, and power encode tradeoffs

	H_0 true	H_a true
Do not reject H_0	correct	Type II, β
Reject H_0	Type I, α	power, $1-\beta$

- Type I error probability: α.
- Type II error probability: β.
- Power: $1-\beta$.

Power depends on effect size, noise, sample size, the test, and α.

Assumptions are part of the answer

Independence	Observations and errors do not carry unmodeled dependence.
Distribution	The selected normal, t, or other approximation is adequate.
Stable process	The mechanism does not shift between the sample and target.
Legal information	No future values enter a forecast or its features.

When an assumption fails, change the method or narrow the claim.

Regression turns association into a conditional model

We observe an outcome Y and a predictor X:

$$Y = f(X) + \varepsilon.$$

- $f(X)$ is the systematic signal we try to learn.
- ε contains variation the model does not explain.
- Prediction combines the estimated signal with honest uncertainty.

Nowcasting example

Use timely surveillance signals X to estimate current influenza activity Y before the official value is complete.

One predictor gives simple linear regression

For observations $i = 1, \ldots, n$,

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

- Y_i response or dependent variable;
- x_i predictor, explanatory variable, or feature;
- β_0 intercept: mean response when $x = 0$;
- β_1 slope: change in mean response for a one-unit increase in x;
- ε_i unobserved error around the conditional mean.

Linear means linear in the coefficients

Linear models

$$\beta_0 + \beta_1 x$$

$$\beta_0 + \beta_1 x^2$$

$$\beta_0 + \beta_1 \log x$$

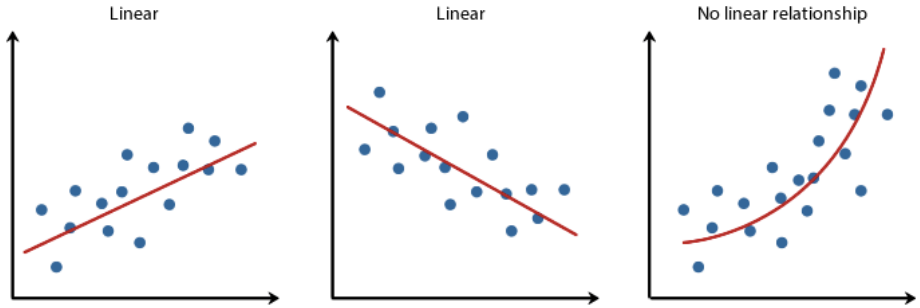
Nonlinear in parameters

$$\beta_0 + e^{\beta_1 x}$$

$$\frac{\beta_0 x}{\beta_1 + x}$$

Transforming the predictor can bend the fitted curve while preserving linearity in the coefficients.

Different patterns require different models



- The first two patterns can be summarized by a straight conditional mean.
- The third pattern has strong association, but a straight line misses its shape.

The fitted line estimates a conditional mean

Population relationship:

$$E(Y_i \mid X_i = x_i) = \beta_0 + \beta_1 x_i.$$

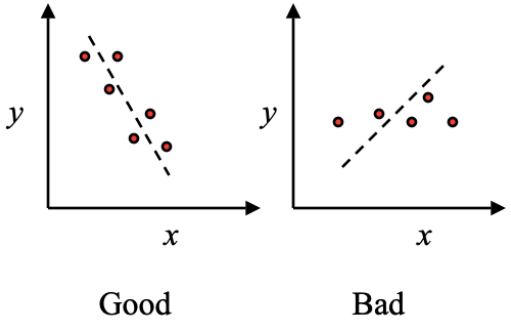
Estimated relationship:

$$\widehat{Y}_i = b_0 + b_1 x_i.$$

Interpretation boundary

The slope describes association in the modeled population. It is not automatically a causal effect, and it may not remain valid outside the observed range.

A best line needs a loss function



A visual judgment is not reproducible.

Least-squares criterion

Choose b_0 and b_1 to minimize the sum of squared vertical residuals.

Squaring prevents positive and negative residuals from canceling and strongly penalizes large misses.

Residuals make fitted error observable

For each observed point,

$$e_i = y_i - \widehat{y}_i = y_i - (b_0 + b_1 x_i).$$

Positive residual

The observation lies above the fitted line. The model underpredicts.

Negative residual

The observation lies below the fitted line. The model overpredicts.

Residuals are observed; model errors are not.

Least squares minimizes squared residuals

$$\mathrm{SSE}(b_0, b_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - (b_0 + b_1 x_i)]^2 .$$

Set the two partial derivatives equal to zero:

$$\frac{\partial \mathrm{SSE}}{\partial b_0} = 0, \qquad \frac{\partial \mathrm{SSE}}{\partial b_1} = 0.$$

Result

The normal equations produce a unique line when the predictor has nonzero variation.

The slope is scaled sample association

Define

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2, \qquad S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}).$$

Then

$$b_1 = \frac{S_{xy}}{S_{xx}} = r\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1\bar{x}.$$

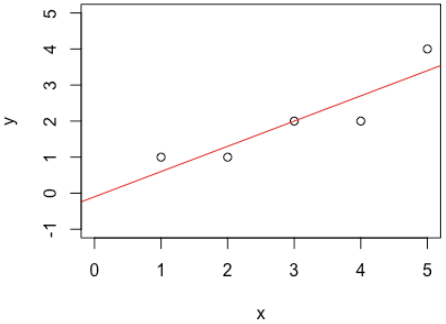
Association sets the direction; the measurement scales set the units.

Work one fit by hand before trusting software

x	1	2	3	4	5
y	1	1	2	2	4

$$\bar{x} = 3, \quad \bar{y} = 2, \quad S_{xx} = 10, \quad S_{xy} = 7.$$

$$b_1 = 0.7, \qquad b_0 = -0.1.$$



Antigravity demo: Appendix Box 1

Interpret the line in units and within its range

The fitted line is

$$\widehat{y} = -0.1 + 0.7x.$$

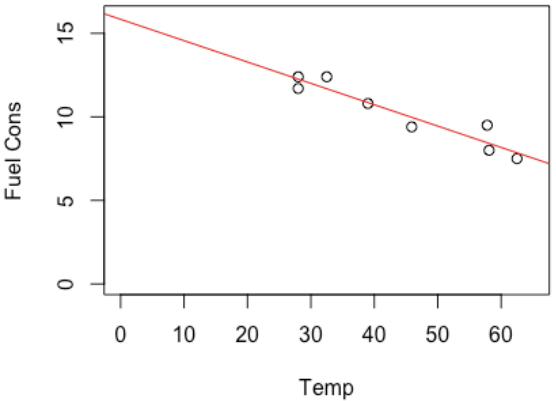
- **Slope:** increasing x by one unit is associated with a 0.7-unit increase in the estimated mean response.
- **Intercept:** the estimated mean response at $x = 0$ is -0.1.
- **Boundary:** $x = 0$ is outside the observed range, so the intercept may have little scientific meaning.
- **Prediction:** at $x = 3$, $\widehat{y} = 2$, but a future observation need not equal 2.

A fuel example connects fit, prediction, and extrapolation

Weekly temperature x and fuel consumption y give

$$\widehat{y} = 15.84 - 0.1279x.$$

- Each 1°F increase is associated with a 0.1279-unit decrease in mean fuel consumption.
- The data cover roughly 28°F to 62.5°F.



Antigravity demo: Appendix Box 2

Residuals estimate errors, but they are not the errors

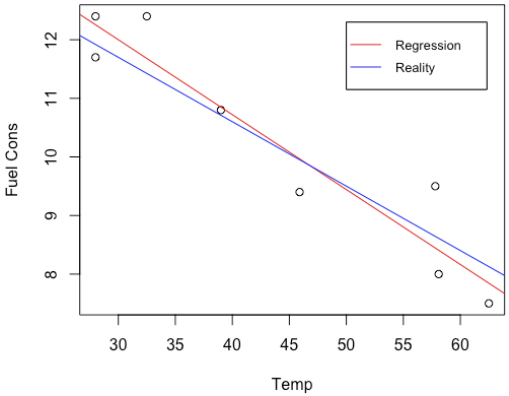
Population model:

$$\varepsilon_i = Y_i - E(Y_i \mid X_i).$$

Fitted sample:

$$e_i = y_i - \widehat{y}_i.$$

- ε_i depends on unknown population coefficients.
- e_i depends on coefficients estimated from the same data.



Least squares creates identities we can audit

With an intercept in the fitted model:

$$\sum_{i=1}^{n} e_i = 0, \qquad \overline{\widehat{y}} = \bar{y}, \qquad \widehat{y}(\bar{x}) = \bar{y}.$$

Fast checks

The fitted line passes through $(\bar{x}, \bar{y})$, and residuals sum to zero up to numerical precision.

Important limitation

These identities can pass even when the model shape, data cutoff, or scientific interpretation is wrong.

Antigravity demo: Appendix Box 3

Audit the implementation, not just the output

```
X = sm.add_constant(x)
fit = sm.OLS(y, X).fit()
y_hat = fit.predict(X)
residual = y - y_hat
```

Independent checks

- reproduce b_0, b_1 by hand;
- check $\sum e_i \approx 0$;
- check the line at $(\bar{x}, \bar{y})$.

Model checks

- inspect the scatterplot;
- verify units and range;
- test data cutoffs and missing values.

Antigravity demo: Appendix Box 4

Form a forecasting team this week

Team structure

- Target: 12 teams of 3 to 4 students.
- Build complementary strengths.
- Rotate data, modeling, operations, and evaluation roles.

Before the next class

- Identify preferred teammates.
- If you need matching, say so early.
- Be ready to work in one reproducible repository.

A team is a forecasting operation, not a division of disconnected tasks.

Exit check: fit, interpret, and audit

For a fitted line $\widehat{y} = 4.2 - 0.6x$:

1. Interpret the slope in context and name the units required.
2. What does the line estimate: an individual outcome or a conditional mean?
3. Name one least-squares identity that should become a code test.
4. What evidence would make you reject the straight-line model?

Next: regression uncertainty and the infectious-disease nowcast kickoff.

Optional: the five-point example exposes every residual

For $\widehat{y} = -0.1 + 0.7x$:

x	y	$\widehat{y}$	$e = y - \widehat{y}$	e^2
1	1	0.6	0.4	0.16
2	1	1.3	−0.3	0.09
3	2	2.0	0.0	0.00
4	2	2.7	−0.7	0.49
5	4	3.4	0.6	0.36
		$\sum$	0.0	1.10

Optional: a small p-value can accompany a small effect

Large data set

A tiny effect can be estimated precisely and produce a small p-value.

Small data set

A practically important effect can remain uncertain and produce a large p-value.

Report together:

- effect estimate and units;
- confidence interval;
- p-value and test definition;
- assumptions and practical consequence.

Appendix Box 1: Inspect and plan

```
The current project contains one file: fuel_consumption.csv.
Do not create or change files yet.

1. Inspect the CSV. Report its columns, number of rows, data types,
   missing values, and observed ranges.
2. Identify the temperature predictor and fuel-consumption response.
3. State the regression estimand and the assumptions needed to
   interpret a simple linear regression.
4. Propose a short, reproducible implementation plan.
5. List numerical and conceptual checks that should be completed
   before the analysis is accepted.

Wait for my approval before creating or editing files.
Do not use the internet or add external data.
```

Appendix Box 2: Implement the regression

```
Proceed with the approved plan.

1. Create analysis.py and parse fuel_consumption.csv.
2. Do not silently remove, replace, or reorder observations.
3. Fit ordinary least squares with temperature as the predictor,
   fuel consumption as the response, and an intercept.
4. Independently calculate:
      b1 = Sxy / Sxx
      b0 = y_bar - b1 * x_bar
5. Verify that the hand calculation and fitted model agree.
6. Calculate fitted values, residuals, SSE, and MSE.
7. Calculate the point prediction at 70 degrees Fahrenheit and
   determine whether it is interpolation or extrapolation.

Run analysis.py and show the numerical results.
Do not use the internet, external data, or an unexplained formula.
```

Appendix Box 3: Audit and test

```
Audit the implementation as a skeptical statistical reviewer.

1. Create test_analysis.py.
2. Test agreement between the formula-based and fitted coefficients.
3. Test that residuals sum to approximately zero.
4. Test that the fitted line passes through (x_bar, y_bar).
5. Test that the mean fitted value equals y_bar.
6. Check that all CSV rows were handled explicitly and that missing
   values, if any, were not silently discarded.
7. Run analysis.py and all tests. Show the commands and results.
8. If a check fails, diagnose the cause before changing the code.

Passing these tests does not prove that a straight-line model is
scientifically adequate. State what the tests do and do not verify.
```

Appendix Box 4: Interpret and report

```
Complete the analysis and produce an evidence-based walkthrough.

1. Create a labeled scatterplot with the fitted regression line.
2. Report the fitted equation with units.
3. Interpret the slope as a change in conditional mean response.
4. Explain whether the intercept has a useful interpretation.
5. Explain the limitation of the prediction at 70 degrees Fahrenheit.
6. Explain why the slope shows association and does not, by itself,
   establish causality.
7. List the created files, numerical results, test evidence,
   assumptions, and unresolved limitations.

Do not claim that successful execution or passing tests proves the
model is correct. Distinguish computation, verification, and judgment.
```